



The Make in India initiative, which seeks to make the country a global manufacturing hub, is expected to be the biggest driver for industrial automation and the IoT to take the big step forward, as global organisations investing in India will no doubt leverage and integrate technology and skills to enhance their competitive edge and capture markets."

— Sudhir Dembi, general manager - plant solutions, Schneider Electric India



including heavily-automated plants like automotive plants, usage of automation as the percentage of the total capital investments has been half of what it is in the West. That means, there are processes in India that we tend not to automate. For instance, you may have a paint shop that is highly automated, but you may not have a fully-automated material-handling system," says Wadhwa. He adds, "The challenge lies in adopting automation as a strategic advantage." So what are those factors that are holding the industry back?

Acceptability among end users.

Getting the end user to invest in something that adds cost to what he or she is doing is a challenge in itself. Gandhi feels that the decision to use automation is generally triggered by a push from the market. He says, "For example, a company may need to launch a new car variant, so there is a push for using automation to go for flexible manufacturing."

Another example is the push for hygiene. "We have customers from the F&B industry who were getting a lot of push from the market to automate the factory where the packaging was being done. While baking has been automated, packaging is being done manually. So they are investing in automating the packaging too keeping the hygiene factor in mind," informs Gandhi.

However, the most important criteria for such decision remain the RoI considering the tune of up-front investment. However, acceptability is going higher off late, which is good for the industry.

Lack of skilled manpower. This

is perhaps the biggest challenge facing the industry and to some extent is also responsible for the difficulty for end users to use automation. Wadhwa points out that lack of skilled people who can engineer, build and commission efficient and holistic automation solutions is a worrying factor.

The fact that these industries are spread in far-flung areas of the country further adds to the challenge of finding manpower locally to deploy automation and in some cases even maintain or operate the systems, adds Gandhi.

Wadhwa feels that it is imperative that the system integrators possess domain knowledge besides having expertise in installing an automation system. This is where the main challenge lies, as this domain knowledge, *per se*, lies mostly with the process equipment or capital goods suppliers and the end users.

"If a person with knowledge of computer programming and programmable logic controller (PLC) commissioning wants to become a system integrator, he or she has to decide the applications, the area of focus, while the challenge for a person with domain knowledge is in combining that strength with an understanding of automation tech-

nologies," explains Wadhwa.

The industry association as well as the automation system providers are providing training with a conscious shift in skill-building now being made towards generating application-specific solutions. Gandhi adds, "We have invested heavily on training. We have training centres in Bengaluru, Mumbai and Gurgaon. Besides this, we also impart e-learning training, which is available to our channel partners or our customers."

Market trends

Automation is not seen as a black box any more but as a piece of intelligence that is captured within machines. And human beings need to be able to interact with this intelligence. So evolution of smart human machine interface (HMI) is seeing a big shift.

Wadhwa opines, "The design of HMI is becoming really important, particularly in terms of how the data is presented. For example, a leading global company in the food-processing area has a project to standardise the look and feel of information across all HMIs at its plants, not only in India but across the globe. This is to enhance comfort among managers/operators/supervisors when they move across locations, from one plant to another."

Another trend is the use of vision systems in quality inspection, informs Gandhi. However, the most interesting one is related to the use of Big Data in manufacturing processes. Gandhi opines that proper application of Big Data in manufacturing can lead to significant improvement in productivity as well as reduction



RoI is of course one basic criterion. Usually the decision to use automation is taken when the end user gets a push from the market. For example, a company may need to launch a new car variant, so there is a push for using automation to go for flexible manufacturing. Another example of this kind of push is the push for hygiene." — Sameer Gandhi, managing director, Omron Automation India

